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Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-8)

ADVANCED INSTRUMENTATION TECHNIQUES

Subject Code : BP811ET

M.Code : 79774

Date of Examination : 14-12-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

- SECTION-A

1. **Write briefly :**

   a) A $-CH_3$ group is detected as a sharp signal but a $-NH_2$ group is detected as a broad signal in H-NMR spectrum of a compound. Why?

   b) What is a metastable ion?

   c) What is the principal difference between DTA and DSC?

   d) What are Bremsstrahlung interactions?

   e) Name the standards used for calibration of IR and UV-VIS spectrophotometers.

   f) Give any two elements used in Radioimmuno assays with their specific applications.

   g) Define partition coefficient.

   h) What is tandem mass spectrometry?

   i) What are the characteristic signals for a $-OCH_2CH_3$ group in H-NMR spectrum?

   j) What do you understand by Index of Hydrogen Deficiency (IHD). Give an example.

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SECTION-B

2. What is the importance of extraction process in pharmaceuticals? Give a detailed account on solid phase extraction.

3. Define the term Chemical shift. What is its advantage over the frequency scale? Describe various factors affecting chemical shift of protons with examples.

4. What is the principle of X-ray spectroscopy? Write a detailed description of various X-ray crystallographic methods.

SECTION-C

5. What are various ionization techniques used in mass spectrometry? Write a detailed account on any one such technique.

6. Differentiate between o-Nitrophenol and *p*-Nitrophenol on the basis of NMR spectra.

7. Predict the mass spectrum of ethyl benzoate in EI mode.

8. What is the principle of TGA? Discuss its application taking the example of Calcium.

9. Write an account on ICH guidelines for method validation.

10. Elaborate on different methods of RIA.

11. What is the principle of LC-MS/MS. Give a descriptive diagram of a LC-MS/MS instrument.

12. Give an explanatory note on variables responsible for accuracy and precision of DTA.

13. Give a step-by-step procedure for calibration of Flame photometer.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

Gian College of Pharmacy

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